

ORIGINAL ARTICLE



Materials and device designs for thermophotovoltaic power conversion

Patrick J. McCann | Majed Khodr

School of Electrical and Computer Engineering, Gallogly College of Engineering, University of Oklahoma, Norman, Oklahoma, USA

Correspondence

Patrick J. McCann, School of Electrical and Computer Engineering, Gallogly College of Engineering, University of Oklahoma, 110 West Boyd St, Norman, OK, 73019-1102, USA.
Email: pmccann@ou.edu

Abstract

Two IV–VI semiconductor alloys, $\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}$ and $\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}$, are proposed for use in designing multiple quantum well (MQW) materials and devices for thermophotovoltaic (TPV) power conversion. These materials can be epitaxially grown on silicon substrates, so they offer the potential for a low cost TPV device manufacturing technology. MQW materials examples are provided for fabricating triple junction TPV devices for power conversion with a 1400°C radiator. Optimal n-type and p-type layer thicknesses for each junction were determined using internal quantum efficiency expressions derived assuming all photogenerated charge collection is by diffusion. Depending on MQW material quality and assumed optical absorption levels, predicted power generation for current matched triple junction devices ranged from 1.7 to 4.6 W/cm² and power conversion efficiencies ranged from 15.9% to 35.7%. These device performance parameters were used in a levelized cost of energy (LCOE) calculation to predict the cost of energy provided by TPV devices with a molten silicon thermal energy storage system. Results show that even for TPV devices fabricated from low quality materials, such energy storage systems will have a low enough LCOE to be competitive with other forms of energy storage and power generation.

KEYWORDS

epitaxial materials on silicon growth substrates, multiple quantum wells, photovoltaics, power generation, thermal energy storage

1 | INTRODUCTION

Thermal energy storage offers a low cost option for long duration energy storage at utility scale. An attractive thermal energy storage medium is molten silicon.¹ Silicon has an unusually large latent heat of fusion, 0.497 kWh/kg, at its 1414°C melting point and is among the most Earth abundant elements. Silicon offers both high energy storage density and low cost,

and technologies for handling its molten form are well developed. The key challenge in developing useful thermal energy storage systems is having an effective way to convert stored thermal energy to electricity. In this paper we describe a materials technology that can be used to fabricate multijunction thermophotovoltaic (TPV) devices.² We show that multiple quantum well (MQW) materials composed of pseudomorphically strained $\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}$ and $\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}$ layers can be

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2023 The Authors. *Energy Science & Engineering* published by Society of Chemical Industry and John Wiley & Sons Ltd.

used to fabricate multijunction TPV devices for conversion of thermal radiation to electrical power. We provide materials and device design models for fabricating multijunction TPV devices where the bandgap energy and layer thickness can be optimized for TPV power conversion applications. These IV–VI semiconductor materials can be grown as epitaxial layers on silicon substrates, which allows for low cost manufacturing. Based on known materials quality parameters for IV–VI semiconductor materials grown on silicon substrates, we show that it should be possible to fabricate triple junction TPV devices with optical-to-electrical power conversion efficiencies as high as 35.7% when converting radiation from a 1400°C thermal radiator, as would likely be the case for a molten silicon thermal energy storage system. A cost model for energy dispatched by a TPV-based thermal energy storage system is also provided, and it shows that the leveled cost of energy (LCOE) from a molten silicon thermal energy storage system can be below \$0.10/kWh. The TPV materials and device technologies described in this paper offer a pathway for development of low cost energy storage technology. With no moving parts, other than pumps for circulating heat exchange fluids, TPV-based thermal energy storage systems will operate much like lithium ion batteries, but their cost will be much lower.

This paper begins with a materials design model involving use of alternating layers of $\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}$ well and $\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}$ barrier alloys that comprise a MQW material. This model provides parameters for growth of MQW materials with bandgap energies between 400 and 700 meV. Analytical expressions are then derived for calculating the internal quantum efficiency (IQE) for electron and hole collection for a given material quality and layer thickness. An example material design is included for a triple junction TPV device designed for power conversion with a 1400°C thermal radiator. Optimal layer thicknesses are provided, and power generation densities and power conversion efficiencies are predicted. These predictions are then used in a cost model to estimate the LCOE for a molten silicon thermal storage system with IV–VI semiconductor TPV devices for power conversion.

2 | IV–VI SEMICONDUCTORS

Unlike III–V semiconductors, IV–VI semiconductors have a strain relaxation mechanism that allows the growth of high crystalline quality epitaxial layers on silicon substrates. Misfit dislocations in IV–VI semiconductor crystals can move easily in the $\langle 110 \rangle$

directions along the $\{100\}$ planes.³ The $\{100\}$ planes in (111)-oriented thin films are at inclined angles, so dislocations created due to temperature cycling with a thermal expansion mismatched substrate will be able to move completely out of the layer, which is observed as triangular crosshatching on epilayer surfaces following cool down from 300°C to 360°C molecular beam epitaxial (MBE) growth temperatures.⁴ Measurement of narrow X-ray diffraction peak widths following more than 1000 temperature cycles from room temperature to 77 K⁵ further show that these thin films are highly tolerant to thermal stresses even at low temperatures. III–V semiconductors do not have this ability to deform plastically without leaving crystalline damage. Thermal expansion matched III–V semiconductor substrates, which are significantly more costly than silicon, are therefore required for III–V semiconductor device fabrication. Procedures for obtaining high crystalline quality IV–VI semiconductor materials on silicon substrates using MBE growth methods have been reported.^{6–9} Materials characterization shows that MBE-grown IV–VI semiconductor MQW materials on silicon can have high charge carrier mobilities and high photoluminescence (PL) intensities. Successful fabrication of mid-IR focal plane arrays¹⁰ and mid-IR vertical cavity surface emitting lasers¹¹ from IV–VI semiconductor layers grown on silicon demonstrate the viability of this materials technology for optoelectronic device fabrication.

Figure 1 is a bandgap versus lattice parameter plot for PbSe-based narrow bandgap IV–VI semiconductor materials. The wide solid solubility range of tin and strontium in PbSe allows growth of $\text{Pb}_{1-x}\text{Sn}_x\text{Se}$ and $\text{Pb}_{1-x}\text{Sr}_x\text{Se}$ ternary alloys, which can comprise the well and barrier materials, respectively, of a MQW material.

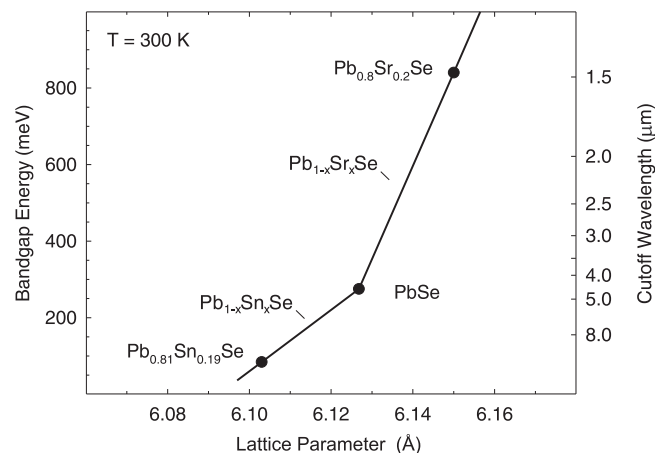


FIGURE 1 Bandgap versus lattice parameter plot for binary PbSe and $\text{Pb}_{1-x}\text{Sn}_x\text{Se}$ and $\text{Pb}_{1-x}\text{Sr}_x\text{Se}$ ternary alloys.

TABLE 1 Parameters for determining strain modified bandgap energies for $\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}$ and $\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}$ layers in a pseudomorphically strained MQW material at 300 K.

	Bulk material properties				Strained MQW material			
	a_o (Å)	D_d (eV)	D_u (eV)	E_g (meV)	a_o (Å)	$\varepsilon_{ }$ (%)	$\varepsilon_{\perp}$ (%)	E_g (meV)
$\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}$	6.103	5.3	−0.2	84	6.127	0.395	−0.961	77
$\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}$	6.151	5.3	−0.2	840	6.127	−0.395	0.961	848

Two ternary alloys are indicated in Figure 1, $\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}$ and $\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}$. The critical layer thickness at which misfit dislocations begin to form due to the 0.79% lattice mismatch between these two alloys, is in the range of 17.5 nm.¹² Strain-balanced MQW materials, with the same 6.127 Å lattice parameter, can thus be obtained if layer thicknesses are below this critical layer thickness. The bandgap energies of the well and barrier materials will be affected by the alternating tensile and compressive strains. The change in bandgap energy is given by Equation (1),

$$\delta E_g = D_d(2\varepsilon_{||} + \varepsilon_{\perp}) + D_u\varepsilon_{\perp}, \quad (1)$$

where D_d and D_u are the dilatation and uniaxial acoustic deformation potentials, respectively, for PbSe and $\varepsilon_{||}$ and $\varepsilon_{\perp}$ are the in-plane and out-of-plane strains.¹³ Table 1 summarizes the effect of this strain, where the room temperature bandgap energies of the strained well and barrier layers are provided. The deformation potentials for PbSe are from Zasavitskii et al.,¹³ and the bandgap energies for unstrained $\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}$ and $\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}$ are from Preier¹⁴ and Shen et al.,¹⁵ respectively. Well material bandgap energy is reduced by in-plane tensile layer strain from 84 to 77 meV, while the barrier material bandgap energy is increased from 840 to 848 meV.

The band structure parameters in the form of effective masses for electrons and holes in IV–VI semiconductors show that the conduction and valance bands are nearly mirror images of each other near the bandgap at the fourfold degenerate L-point. Based on this mirror symmetry, we assume electron and hole effective masses are equal and band edge discontinuities at the $\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}/\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}$ interfaces are symmetric. Figure 2 shows the band diagram for a $\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}/\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}/\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}$ MQW material where conduction and valence band well depths are 385 meV. This well depth, along with the transverse and longitudinal masses for electrons and holes in $\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}$ ¹⁴ are used with Schrödinger's equation to calculate MQW bandgap energies as a function of well width. The [111] direction of quantum confinement creates two different effective masses for electrons and holes because one of the four L-valleys

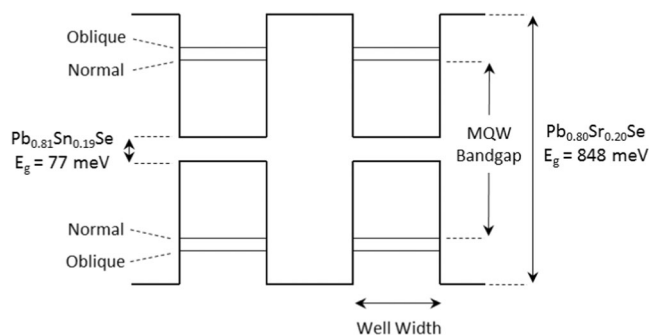


FIGURE 2 Band diagram for a $\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}/\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}/\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}$ MQW material. The bottoms of the normal and oblique valley subbands are indicated.

lies along the direction of quantum confinement and the other three lie at oblique angles.¹⁶ MQW materials will thus have a band structure composed of a lower energy normal valley subband and three degenerate oblique valley subbands at a higher energy as indicated in Figure 2. With this subband structure, the effective density of states will be smaller than that for a bulk material with a similar bandgap energy, an effect that reduces reverse bias saturation current and increases open circuit voltage.¹⁷ Figure 3 shows interband transition energies in $\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}/\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}/\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}$ MQW materials plotted as a function of $\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}$ well width. A nearly linear decrease in direct normal valley interband transition energy from about 700 meV at a well width of 1.4 nm to below 400 meV at a width of 4.2 nm is shown.

In addition to enabling a more than 30% improvement in power generation density,¹⁷ which makes IV–VI semiconductor TPV devices comparable in performance to III–V semiconductor TPV devices, a MQW materials design is also attractive for TPV device manufacturing. Rather than being controlled by alloy composition, the bandgap energy is instead determined by quantum well width, which in turn is determined by the timing of shutter openings and closings during growth of the $\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}$ well layer. Only four effusion cells are needed for growing these MQW materials, PbSe, SnSe, Sr, and Se, where Se flux is used to control epilayer stoichiometry, and all effusion cells remain at stable

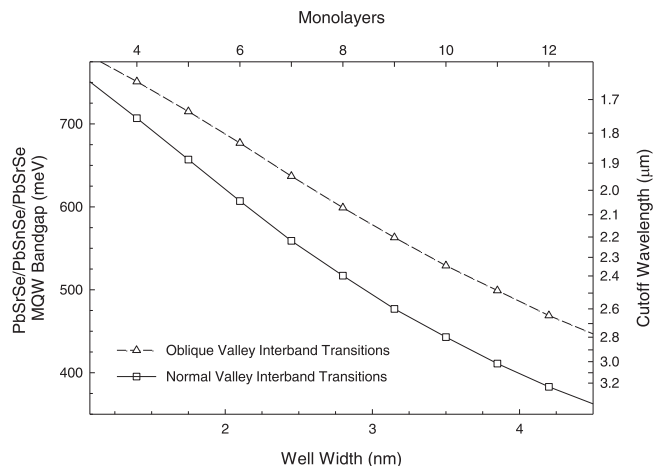


FIGURE 3 Normal and oblique valley interband transition energies as a function of $\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}$ well width.

temperatures with constant fluxes during growth of different bandgap materials. A complete multijunction TPV device structure can therefore be grown with stable MBE growth chamber conditions, a feature that helps to achieve consistently high quality epitaxial layer growth. Ability to obtain a wide range of bandgap energies with the same MBE growth conditions will provide important flexibility in optimizing TPV device designs for different applications. For example, TPV devices for power conversion from relatively low radiator temperatures can be made with relatively wide quantum wells and long shutter open times, while devices for matching the higher energy photon flux from a relatively hot radiator can be made using shorter shutter open times. Such manufacturing flexibility provides many options for the commercialization of TPV devices for a range of power conversion applications, a benefit that can expedite manufacturing scale-up and TPV device cost reduction.

Layers with n-type and p-type conductivity can be obtained with the inclusion of two dopant cells in the MBE growth chamber. Bi_2Se_3 for n-type doping and Tl_2Se for p-type doping have been demonstrated.¹⁸ Silver¹⁹ and sodium²⁰ have also been used for p-type doping. Hall effect measurements show that impurity doping in IV–VI semiconductors can give electron and hole majority carrier densities between 2×10^{17} and $8 \times 10^{18} \text{ cm}^{-3}$. Controlling electron and hole concentrations below about $1 \times 10^{17} \text{ cm}^{-3}$ is difficult with these materials due to the wide solid solubility range of anion and cation vacancies, which act as donors and acceptors, respectively, in IV–VI semiconductor materials.²¹ Degenerate doping, which is necessary for forming tunnel junctions and obtaining series connected multijunction device structures, typically occurs at carrier densities above $3 \times 10^{18} \text{ cm}^{-3}$ in these materials,²² which is within

the controllable doping range. MBE technology is therefore available for the growth of broad spectral coverage multijunction device structures with bandgap energies that are well matched to thermal radiation spectra.

3 | TPV MATERIALS DESIGN

A cross-sectional drawing of a triple junction TPV device structure with thermal radiation incident on the top junction is shown in Figure 4. The bandgap of each junction is selected to match a portion of the emission spectrum of a thermal radiator with the top junction having the largest bandgap energy and the bottom junction having the smallest. Figure 5 provides a

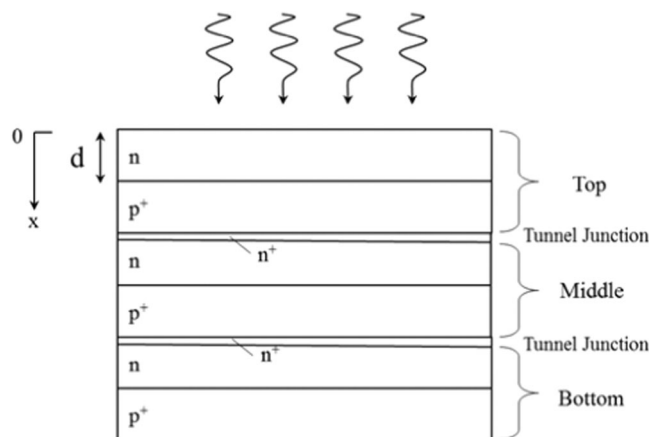


FIGURE 4 Triple junction thermophotovoltaic device structure.

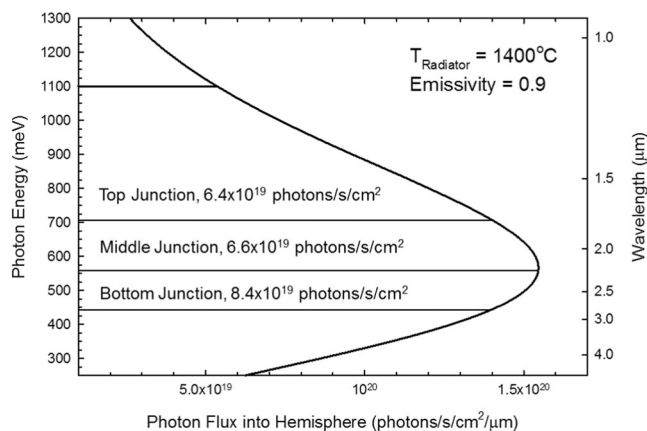


FIGURE 5 Portions of the thermal emission spectrum that can be collected with a triple junction thermophotovoltaic device with bandgap energies of 707, 559, and 443 meV. The number of absorbable photons in each junction is indicated for a 1400°C radiator with an emissivity of 0.9.

specific example for a triple junction TPV device with materials designed for absorption of radiation from a 1400°C radiator. Three spectral bands with approximately equal photon fluxes are obtained with top, middle, and bottom junction bandgap energies of 707, 559, and 443 meV, respectively. These bandgap energies can be obtained with $\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}/\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}/\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}$ MQW materials with well widths of 1.4, 2.45, and 3.5 nm, respectively. Assuming an upper limit of 1.1 eV for photon absorption by the top junction, these three junctions are illuminated with a total of 21.7 W/cm^2 of absorbable optical power, which is 54% of the total radiant optical power from a 1400°C radiator.

As shown in Figure 4, each junction has a lightly doped n-type side on a heavily doped p^+ side, which also serves as the p^+ side of two tunnel junctions. Two thin, heavily doped n^+ layers serve as the n^+ sides of the tunnel junctions, which provide a series connection of the three junctions through the middle junction. Table 2 provides materials parameters for the top, middle, and bottom junctions, each with a lightly doped n-type side and the more heavily doped p^+ side. Table 2 includes a calculation of the intrinsic charge carrier density for MQW materials that comprise each junction, $n_i = N_{\text{MQW}}^{\text{Normal}} e^{qE_N/2kT} + N_{\text{MQW}}^{\text{Oblique}} e^{qE_O/2kT}$, where E_N and E_O are the bandgap energies between the normal and oblique valley subbands, respectively. The effective density of states for each subband is calculated using $N_{\text{MQW}} = MmkT/2d_{\text{QW}}\pi\hbar^2$, where M is the valley degeneracy ($M=1$ for the normal valley and $M=3$ for the oblique valleys), $2d_{\text{QW}}$ is MQW period width, which is twice the well width for a strain balanced material with the same $\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}$ barrier width, and m is the charge carrier effective mass. Using equations from Khodr et al.²³ and the transverse and longitudinal masses for charge carriers in $\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}$,¹⁴ the masses for electrons and holes are $0.044 m_0$ in the normal valley and $0.025 m_0$ in the oblique valley, where m_0 is the free electron mass. The built-in potential, $qV_0 = kT \ln \frac{n_n p_p}{n_i^2}$,

and depletion width, $W = \sqrt{\frac{2\epsilon_r \epsilon_0 V_0}{q} \left(\frac{1}{n_n} + \frac{1}{p_p} \right)}$, for each junction are also provided.

4 | ABSORPTION AND CHARGE CARRIER GENERATION

The photogeneration rate for holes and electrons can be obtained from the Beer–Lambert law for photon absorption, $N(x) = N_0 e^{-\alpha x}$, where $N(x)$ is the photon flux as a function of depth, x , into the material, α is the absorption coefficient, and N_0 is the flux of absorbable photons ($N_0 = \text{incident photon flux} - \text{reflected photon flux}$). Assuming photon flux decrease is due to electron-hole pair generation across the semiconductor bandgap, the electron-hole pair generation rate can be expressed as the change in photon flux over distance,

$$G(x) = -\frac{dN}{dx} = N_0 \alpha e^{-\alpha x}. \quad (2)$$

Optical absorption in the MQW material will be determined by the joint density of states associated with direct interband transitions between normal and oblique valley states. All four L-valleys will participate in optical absorption when photon energies exceed the oblique valley bandgap energy, so the joint density of states at the bottom of the oblique valley subband will be similar to that for a chemically similar bulk material. An estimate for MQW absorption coefficient can therefore be made from measured absorption coefficients for PbSe. Figure 6 shows plots of $\alpha e^{-\alpha x}$ for different absorption coefficients corresponding to photon energies that are up to 110 meV above the bandgap energy of PbSe. IV–VI semiconductors have relatively large absorption coefficients due to their multivalley band structure,²⁴ and this is reflected in the large generation rates within the first 2–3 μm of a IV–VI semiconductor material even for near-band edge generation where absorption is weakest.

TABLE 2 MQW materials parameters at room temperature for three different bandgap materials.

	Well (nm)	MQW E_g (meV)	n_i (cm^{-3})	n_n (cm^{-3})	p_p (cm^{-3})	qV_0 (meV)	W (nm)
Top	1.40	707	3.4×10^{12}	3×10^{17}	3×10^{18}	652	78
Middle	2.45	559	2.7×10^{13}	3×10^{17}	3×10^{18}	544	71
Bottom	3.50	443	1.7×10^{14}	3×10^{17}	3×10^{18}	449	65

Note: Electron densities of $3 \times 10^{17} \text{ cm}^{-3}$ and hole densities of $3 \times 10^{18} \text{ cm}^{-3}$ are assumed for the n-type and p^+ -type sides of each junction, respectively. A relative permittivity of 22.9 is assumed for depletion width calculation.

Table 3 lists the absorption band width for the top, middle, and bottom junctions described in Table 2. The middle and bottom junctions have spectral bands defined by their bandgaps and the bandgap(s) of the layer(s) above. In the case of the top junction, our model assumes zero absorption of photons at energies above 1100 meV, an energy range where the photon flux from a 1400°C radiator is relatively low and decreasing. Table 3 also lists the bandgap energy difference between normal and oblique valley direct interband transitions, the median photon energy above the bandgap in each spectral band for a 1400°C radiator, and the absorption coefficient of PbSe at the same above bandgap photon energy. The top junction has the broadest absorption band with a median photon energy of 903 meV, 196 meV above the MQW bandgap energy, where the representative absorption coefficient is $1.34 \times 10^4 \text{ cm}^{-1}$ according to Figure 6. With more narrow absorption bands and lower median photon energies above the normal valley bandgap, the middle and bottom junctions

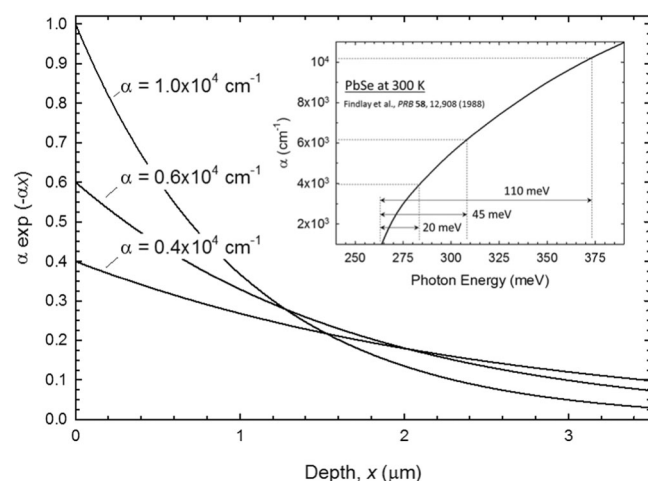


FIGURE 6 Plots of $\alpha e^{-\alpha x}$ versus depth, x , for different absorption coefficients. The inset shows measured PbSe absorption coefficient as a function of photon energy.²⁴ Representative absorption coefficients for the above bandgap photon energies are indicated.

have lower representative absorption coefficients of 0.82×10^4 and $0.72 \times 10^4 \text{ cm}^{-1}$, respectively.

Estimates for absorbable photon flux by each MQW material, N_o , are provided in Table 4. Nonoptimized and optimized optical coatings are considered. The amount of optical absorption by a IV–VI semiconductor material can be estimated using the Fresnel equations, photon incidence angle, and index of refraction. Assuming an index of refraction of 4.5 for the top junction material,²⁵ the amount of light transmitted into the MQW material is about 60% up to an incidence angle of 75° from the normal. Absorption decreases dramatically for glancing incident photons, falling to below 45% for an incidence angle of 85°. A relatively large fraction of the photon flux from a planar radiator will be at these larger incident angles, so to account for this effect, which has been observed in TPV generator design,²⁶ we will assume 50% optical absorption for TPV devices fabricated without optimized optical coatings.

Optimization of optical coatings on both the TPV device and the radiator can enhance optical absorption. For example, PbSe mid-IR detectors fabricated with CaF_2 antireflective coatings exhibited more than 75% absorption across spectral bandwidths similar to those for the junctions modeled here.²⁷ Implementing photonic bandgap²⁸ and meta-optics²⁹ technologies, for example, can further enhance absorption by shaping, both spectrally and spatially, optical emission from a thermal radiator to match the absorption spectrum and geometry of the TPV device. For optimized optical coatings, absorption is assumed to increase to 60% for the top junction and to 80% for the middle and bottom junctions, where the more narrow and well defined absorption bands should allow more effective optical engineering. The incident photon flux for each junction is calculated using Planck's radiation law and assumes a 1400°C thermal radiator with an emissivity of 0.9, as would likely be the case for a molten silicon storage system and a thermal radiator fabricated from high emissivity silicon carbide materials.

TABLE 3 Absorption band width, bandgap difference between normal and oblique valleys, median photon energy above MQW bandgap for radiation from a 1400°C thermal radiator, and representative absorption coefficients for each junction.

Junction	Absorption band width (meV)	Normal-oblique valley separation (meV)	Median photon energy above MQW Bandgap (meV)	PbSe absorption coefficient at median photon energy above bandgap (cm^{-1})
Top	393	44	196	1.34×10^4
Middle	148	78	73	0.82×10^4
Bottom	116	86	59	0.72×10^4

TABLE 4 Estimates for absorbable photon fluxes from a 1400°C radiator with a 0.9 emissivity in the top, middle, and bottom junctions for a triple junction TPV device.

Junction	1400°C 0.9 emissivity photon flux (photons/s/cm ²)	No optical coatings		Optimized optical coatings	
		Assumed absorption (%)	Absorbable photon flux, N_o (photons/s/cm ²)	Assumed absorption (%)	Absorbable photon flux, N_o (photons/s/cm ²)
Top	6.4×10^{19}	50	3.2×10^{19}	60	3.8×10^{19}
Middle	6.6×10^{19}	50	3.3×10^{19}	80	5.3×10^{19}
Bottom	8.4×10^{19}	50	4.2×10^{19}	80	6.7×10^{19}

Note: Values for devices without and with optimized optical coatings are listed.

5 | DIFFUSION COLLECTION MODEL

The charge generation curves shown in Figure 6 suggest that optimal junction thicknesses for PbSe-based TPV devices will be in the range of 2–3 μm . Photogenerated electrons and holes contribute to TPV device photocurrents when they drift through the depletion region. Photogenerated electrons and holes generated outside of the depletion region will diffuse until they either reach the depletion region edge and drift through it or they recombine before reaching the depletion region. Electrons and holes generated inside the depletion region will immediately drift and thus will all be collected, giving an IQE of 100% for this region of the junction material. However, depletion widths are less than 80 nm in these materials, so most photogeneration will be outside of the depletion region. A model for photocurrent calculation can be developed by assuming all charge collection is by diffusion of minority charge carriers to the metallurgical junction. In this model, the depletion width is assumed to be zero, and all charge collection occurs at $x = d$, as shown in Figure 4 for the top junction. This assumption, which introduces only a small underestimate of photocurrent since it neglects the advantage of 100% IQE for charge generated inside a depletion region, allows the derivation of analytical expressions for short circuit photocurrent as a function of materials and device design parameters. Note that this diffusion-only charge collection model can be used in general to analyze IV–VI semiconductor TPV devices made from either MQW or bulk materials.

5.1 | Hole collection from n-type top layer

A collection of holes generated in the n-type top layer, which has a thickness d , as shown in Figure 4, occurs

when they diffuse through the layer and reach the junction where they drift into the p-type bottom layer and contribute to photocurrent. Hole collection probability, $c(x)$, as a function of depth, x , into the layer is obtained from the minority carrier continuity equation³⁰

$$c(x) = \frac{e^{x/L_p} + e^{-x/L_p}}{e^{d/L_p} + e^{-d/L_p}}, \quad (3)$$

$$L_p = \sqrt{D_p \tau_p}, \quad (4)$$

where L_p , D_p , and τ_p are the hole diffusion length, hole diffusion coefficient, and hole lifetime, respectively. Equation (3) assumes that surface recombination velocity is negligible, less than about 5000 cm/s. The fraction of holes that are able to diffuse and become collected by the p-type side increases as x increases, reaching 100% for holes generated at the junction where $x = d$, the n-type layer thickness. Hole collection rate can be expressed as

$$\frac{dp_n}{dt} = \int_0^d G(x)c(x)dx. \quad (5)$$

Substituting the hole generation rate, Equation (2), and the hole collection probability, Equation (3), into Equation (5) gives

$$\frac{dp_n}{dt} = \int_0^d N_o \alpha e^{-\alpha x} \frac{e^{x/L_p} + e^{-x/L_p}}{e^{d/L_p} + e^{-d/L_p}} dx \quad (6)$$

and after integrating

$$\frac{dp_n}{dt} = \frac{N_o \alpha L_p}{e^{d/L_p} + e^{-d/L_p}} \left[\frac{e^{(d/L_p - \alpha d)} - 1}{1 - \alpha L_p} - \frac{e^{-(d/L_p + \alpha d)} - 1}{1 + \alpha L_p} \right]. \quad (7)$$

This rate of hole collection divided by the rate of absorbed photons, N_o , is the internal quantum efficiency

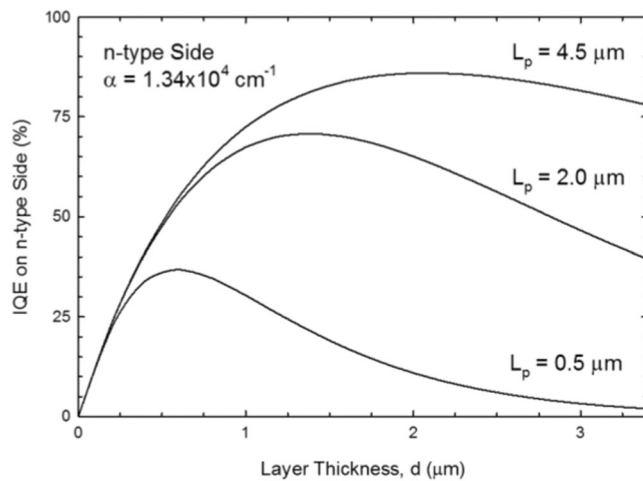


FIGURE 7 IQE_h versus n-type layer thickness for different hole diffusion lengths in a high absorption ($\alpha = 1.34 \times 10^4 \text{ cm}^{-1}$) material.

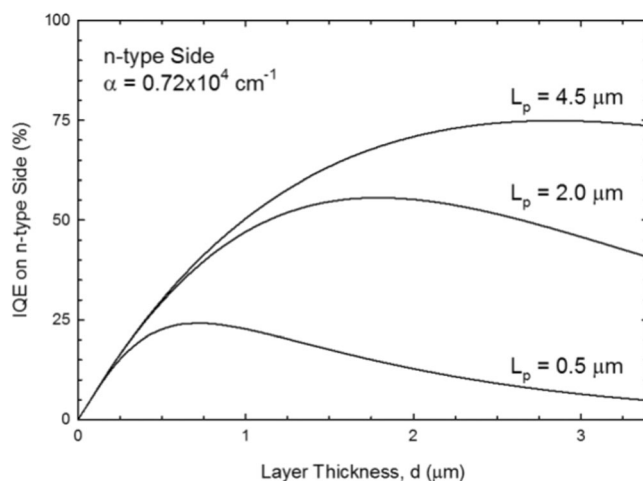


FIGURE 8 IQE_h versus n-type layer thickness for different hole diffusion lengths in low absorption ($\alpha = 0.72 \times 10^4 \text{ cm}^{-1}$) material.

for the collection of holes (IQE_h) generated in the n-type top layer

$$\text{IQE}_h = \frac{dp_n}{N_0 dt} = \frac{\alpha L_p}{e^{d/L_p} + e^{-d/L_p}} \left[\frac{e^{(d/L_p - \alpha d)} - 1}{1 - \alpha L_p} - \frac{e^{-(d/L_p + \alpha d)} - 1}{1 + \alpha L_p} \right] \quad (8)$$

Figure 7 shows IQE_h values obtained using Equation (8) and an absorption coefficient of $1.34 \times 10^4 \text{ cm}^{-1}$ as a function of n-type top layer thickness for three different hole diffusion lengths. Optimal IQE_h values range from 37% for a low quality material with a hole diffusion length of $0.5 \mu\text{m}$ to 86% for a high quality material with a hole diffusion length of $4.5 \mu\text{m}$. Figure 8 shows the calculated IQE_h values for a material with an absorption coefficient of $0.72 \times 10^4 \text{ cm}^{-1}$. Optimal IQE_h values are slightly lower, ranging from 24% for a low quality material to 75% for a high quality material. Optimal layer thicknesses for low quality material range between 0.60 and $0.73 \mu\text{m}$ depending on absorption coefficient. High quality material, which can have two to three times larger IQE_h values, will have optimal layer thicknesses between 2.1 and $2.8 \mu\text{m}$. Table 5 summarizes IQE_h and optimal layer thickness values for the top, middle, and bottom junctions for low, medium, and high quality materials with hole diffusion lengths of 0.5 , 2.0 , and $4.5 \mu\text{m}$, respectively.

5.2 | Electron collection from p-type bottom layer

A collection of electrons generated in the p-type bottom layer, which has a thickness d , occurs when they diffuse through the layer and reach the junction where they then drift into the n-type top layer and contribute to photocurrent. Electron collection probability, $c(x)$, as a

TABLE 5 Optimal n-type top layer thicknesses, d_t , and corresponding IQE_h values for different quality n-type top layers in the top, middle, and bottom junctions of a triple junction TPV device as depicted in Figures 4 and 5.

Junction	Low quality, $L_p = 0.5 \mu\text{m}$		Medium quality, $L_p = 2.0 \mu\text{m}$		High quality, $L_p = 4.5 \mu\text{m}$	
	Optimal thickness, d_t (μm)	IQE_h (%)	Optimal thickness, d_t (μm)	IQE_h (%)	Optimal thickness, d_t (μm)	IQE_h (%)
Top	0.60	37	1.4	71	2.1	86
Middle	0.68	27	1.7	59	2.7	78
Bottom	0.73	24	1.8	56	2.8	75

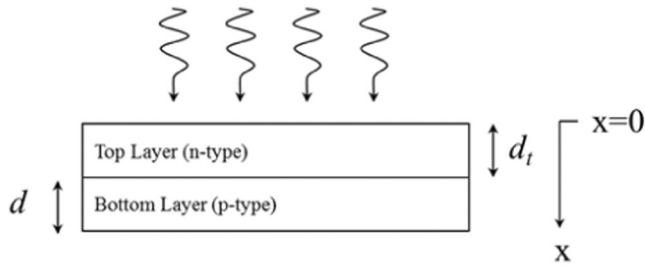


FIGURE 9 Junction parameters for modeling electron collection from the p-type bottom layer with thickness d and n-type top layer thickness d_t .

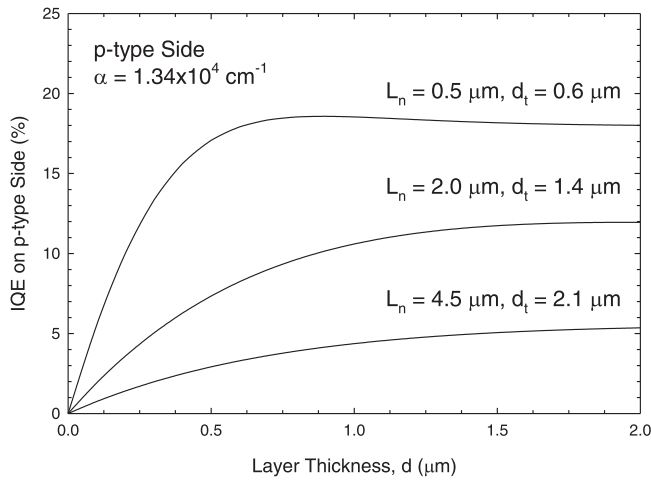


FIGURE 10 IQE_e as a function of p-type layer thickness for different electron diffusion lengths in a high absorption ($\alpha = 1.34 \times 10^4 \text{ cm}^{-1}$) material.

function of depth, x , into the layer is obtained from the minority carrier continuity equation³⁰

$$c(x) = \frac{e^{d_t/L_n} e^{(d-x)/L_n} + e^{-d_t/L_n} e^{-(d-x)/L_n}}{e^{d/L_n} + e^{-d/L_n}}, \quad (9)$$

$$L_n = \sqrt{D_n \tau_n}, \quad (10)$$

where L_n , D_n , τ_n , and d_t are the electron diffusion length, electron diffusion coefficient, electron lifetime, and top layer thickness, respectively. The fraction of electrons that are able to diffuse in the negative x direction and become collected by the n-type side at $x = d_t$ increases as x decreases, reaching 100% at $x = d_t$. The rate of electron collection from the p-type bottom layer as shown in Figure 9 can be expressed as

$$\frac{dn_p}{dt} = \int_{d_t}^{d+d_t} G(x)c(x)dx. \quad (11)$$

Substituting the electron generation rate, Equation (2), and the electron collection probability, Equation (9), gives

$$\frac{dn_p}{dt} = \int_{d_t}^{d+d_t} N_0 \alpha \frac{e^{-\alpha x} \frac{e^{d_t/L_n} e^{(d-x)/L_n} + e^{-d_t/L_n} e^{-(d-x)/L_n}}{e^{d/L_n} + e^{-d/L_n}} dx}{e^{d/L_n} + e^{-d/L_n}} \quad (12)$$

and after integrating

$$\frac{dn_p}{dt} = \frac{N_0 \alpha}{e^{d/L_n} + e^{-d/L_n}} \left[\frac{e^{-(d+d_t)/L_n}}{(1/L_n - \alpha)} \left(e^{(1/L_n - \alpha)(d+d_t)} - e^{(1/L_n - \alpha)d_t} \right) - \frac{e^{(d+d_t)/L_n}}{(1/L_n + \alpha)} \left(e^{-(1/L_n + \alpha)(d+d_t)} - e^{-(1/L_n + \alpha)d_t} \right) \right]. \quad (13)$$

This rate of electron collection divided by the rate of absorbed photons, N_0 , gives the internal quantum efficiency for the collection of electrons (IQE_e) generated in the p-type bottom layer

$$\text{IQE}_e = \frac{dn_p}{N_0 dt} = \frac{\alpha}{e^{d/L_n} + e^{-d/L_n}} \left[\frac{e^{-(d+d_t)/L_n}}{(1/L_n - \alpha)} \left(e^{(1/L_n - \alpha)(d+d_t)} - e^{(1/L_n - \alpha)d_t} \right) - \frac{e^{(d+d_t)/L_n}}{(1/L_n + \alpha)} \left(e^{-(1/L_n + \alpha)(d+d_t)} - e^{-(1/L_n + \alpha)d_t} \right) \right]. \quad (14)$$

Figure 10 shows IQE_e values obtained using Equation (14) for a high absorption coefficient material, $\alpha = 1.34 \times 10^4 \text{ cm}^{-1}$, as would be the case for the top junction material according to data in Table 3. Plots are shown for three different electron diffusion lengths representing different quality materials. Values used for the top layer thicknesses, d_t , are the optimal layer thickness for the n-type top layer as shown in Table 5. The maximum IQE_e for a high-quality material with a $4.5 \mu\text{m}$ electron diffusion length is 5.4%, whereas a low-quality material with a $0.5 \mu\text{m}$ electron diffusion length has a maximum IQE_e of 19%. The influence of the top layer thickness is evident in these data. A lower quality material will have a thinner top layer, allowing the p-type bottom layer to be exposed to a larger portion of absorbable photon flux, and this allows the p-type bottom layer to participate more in overall junction performance. Figure 11 shows IQE_e versus layer thickness for a

low absorption coefficient junction material, $\alpha = 0.72 \times 10^4 \text{ cm}^{-1}$, as would be the case for a bottom junction material. Plots are shown for different electron diffusion lengths where the optimal top layer thicknesses, obtained from Table 5, are used for the d_t values. The maximum IQE_e for a high-quality material with a $4.5 \mu\text{m}$ electron diffusion length is 9.3%, whereas a low-quality material with a $0.5 \mu\text{m}$ electron diffusion length has a maximum IQE_e of 16%. As with the high absorption coefficient material, the top layer thickness has a strong influence on how much charge collection is provided by the bottom p-type layer. Table 6 summarizes optimal layer thickness and IQE_e data for high, medium, and low quality materials.

6 | SHORT CIRCUIT CURRENT

Short circuit current for each junction is the elementary charge times the sum of the hole and electron collection rates, which can be expressed in terms of absorbed

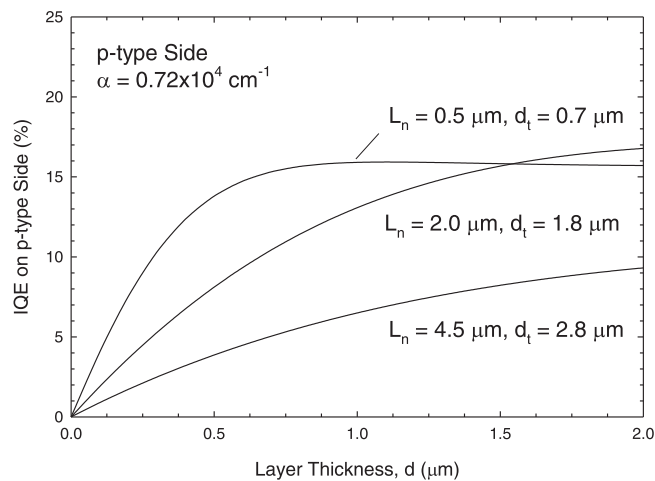


FIGURE 11 IQE_e as a function of p-type layer thickness for different electron diffusion lengths in a low absorption ($\alpha = 0.724 \times 10^4 \text{ cm}^{-1}$) material.

TABLE 6 Optimal p-type bottom layer thicknesses and corresponding IQE_e values for different quality material in the top, middle, and bottom junctions of a triple junction TPV device as depicted in Figures 4 and 5.

Junction	Low quality, $L_n = 0.5 \mu\text{m}$		Medium quality, $L_n = 2.0 \mu\text{m}$		High quality, $L_n = 4.5 \mu\text{m}$	
	Optimal thickness (μm)	IQE_e (%)	Optimal thickness (μm)	IQE_e (%)	Optimal thickness (μm)	IQE_e (%)
Top	0.90	18.6	2.1	11.9	2.9	5.5
Middle	1.10	17.0	2.5	16.4	3.9	9.2
Bottom	1.20	15.9	2.6	17.1	4.2	10.5

Note: These values are for the n-type top layer thicknesses, d_t , shown in Table 5.

photon flux and the internal quantum efficiencies for hole and electron collection,

$$J_{sc} = J_p + J_n = q \frac{dp_n}{dt} + q \frac{dn_p}{dt} = qN_0(\text{IQE}_h + \text{IQE}_e). \quad (15)$$

Absorbed photon flux data, as shown in Table 3, and Equations (8) and (14) can be used to calculate J_{sc} values for triple junction IV–VI semiconductor MQW TPV devices. Optimizing triple junction device performance requires the matching of photogeneration currents for each junction. Layer thickness can be used as an adjustable parameter for achieving current matching conditions. Table 7 provides device design parameters for fabricating current-matched triple junction TPV devices with nonoptimized optical coatings. Table 8 provides the same parameters for devices with optimized optical coatings.

7 | ELECTRIC POWER GENERATION

Electric power delivered to a load by photovoltaic power generation is the product of the pn junction's short circuit current, open circuit voltage, and fill factor (FF)

$$P_{el} = J_{sc} V_{oc} \text{FF}. \quad (16)$$

The open circuit voltage for each junction can be determined from the short circuit current, J_{sc} ,

$$V_{oc} = \frac{kT}{q} \ln \left(\frac{J_{sc}}{J_0} + 1 \right), \quad (17)$$

where J_0 is the reverse bias saturation current,

$$J_0 = n_i^2 \sqrt{\frac{qkT\mu}{\tau}} \left(\frac{1}{n_n} + \frac{1}{p_p} \right). \quad (18)$$

TABLE 7 Layer thicknesses and short circuit currents for current-matched junctions in a triple junction TPV device with nonoptimized optical coatings.

Junction	Layers	Low quality, $L = 0.5 \mu\text{m}$				High quality, $L = 4.5 \mu\text{m}$			
		Total d_t, d (μm)	d_t, d (μm)	J_p, J_n (A/cm^2)	J_{sc} (A/cm^2)	Total d_t, d (μm)	d_t, d (μm)	J_p, J_n (A/cm^2)	J_{sc} (A/cm^2)
Top	n-type	3.36	0.60	1.9	2.3	7.90	2.10	4.4	4.5
	p-type		0.15	0.4			0.20	0.1	
Middle	n-type		0.68	1.4			2.70	4.1	
	p-type		0.90	0.9			1.30	0.4	
Bottom	n-type		0.73	1.6			0.60	2.3	
	p-type		0.30	0.7			1.00	2.2	

TABLE 8 Layer thicknesses and short circuit currents for current-matched junctions in a triple junction TPV device with optimized optical coatings.

Junction	Layers	Low quality, $L = 0.5 \mu\text{m}$				High quality, $L = 4.5 \mu\text{m}$			
		Total d_t, d (μm)	d_t, d (μm)	J_p, J_n (A/cm^2)	J_{sc} (A/cm^2)	Total d_t, d (μm)	d_t, d (μm)	J_p, J_n (A/cm^2)	J_{sc} (A/cm^2)
Top	n-type	3.21	0.60	2.2	3.0	7.15	2.10	5.2	5.3
	p-type		0.30	0.8			0.20	0.1	
Middle	n-type		0.68	1.6			2.70	4.7	
	p-type		0.80	1.4			1.20	0.6	
Bottom	n-type		0.73	2.6			0.60	3.7	
	p-type		0.10	0.4			0.35	1.6	

TABLE 9 Performance parameters for electrical power generation by a triple junction TPV device with a 1400°C thermal radiator.

Junction	Low material quality				High material quality			
	$\mu = 200 \text{ cm}^2/\text{Vs}, \tau = 0.5 \text{ ns}, L = 0.5 \mu\text{m}$				$\mu = 800 \text{ cm}^2/\text{Vs}, \tau = 10 \text{ ns}, L = 4.5 \mu\text{m}$			
	J_{sc} (A/cm^2)	J_o (A/cm^2)	V_{oc} (V)	P_{el} (W/cm^2)	J_{sc} (A/cm^2)	J_o (A/cm^2)	V_{oc} (V)	P_{el} (W/cm^2)
Top	2.3	$6.90\text{E}-7$	0.39	0.78	4.5	$3.09\text{E}-7$	0.43	1.70
Middle		$4.35\text{E}-5$	0.28	0.56		$1.95\text{E}-5$	0.32	1.27
Bottom		$1.72\text{E}-3$	0.19	0.37		$7.71\text{E}-4$	0.22	0.89
Total			0.86	1.72			0.98	3.86

Note: Values are provided for low and high quality epitaxial materials with nonoptimized optical coatings.

Fill factor is calculated using a normalized open circuit voltage according to³¹

$$FF = \frac{V_m - \ln(V_m + 0.72)}{V_m + 1}. \quad (19)$$

where $V_m = qV_{oc}/kT$. This expression assumes parasitic series and shunt resistances are zero and infinite, respectively, and that the pn junction has an ideality

factor of 1. Table 9 provides a summary of predicted TPV device performance parameters for low and high quality materials with nonoptimized optical coatings, and Table 10 provides the same parameters for low and high quality materials with optimized optical coatings. Charge carrier mobility, μ , and minority carrier lifetime, τ , determine minority charge carrier diffusion length,

$$L = \sqrt{\mu\tau kT/q}. \quad (19')$$

TABLE 10 Performance parameters for electrical power generation by a triple junction TPV device with a 1400°C thermal radiator.

Junction	Low material quality				High material quality			
	$\mu_n = 200 \text{ cm}^2/\text{Vs}, \tau = 0.5 \text{ ns}, L = 0.5 \mu\text{m}$				$\mu = 800 \text{ cm}^2/\text{Vs}, \tau = 10 \text{ ns}, L = 4.5 \mu\text{m}$			
	$J_{sc} \text{ (A/cm}^2\text{)}$	$J_o \text{ (A/cm}^2\text{)}$	$V_{oc} \text{ (V)}$	$P_{el} \text{ (W/cm}^2\text{)}$	$J_{sc} \text{ (A/cm}^2\text{)}$	$J_o \text{ (A/cm}^2\text{)}$	$V_{oc} \text{ (V)}$	$P_{el} \text{ (W/cm}^2\text{)}$
Top	3.0	6.90E−7	0.41	1.05	5.3	3.09E−7	0.43	2.02
Middle		4.35E−5	0.29	0.76		1.95E−5	0.32	1.52
Bottom		1.72E−3	0.19	0.51		7.71E−4	0.23	1.07
Total			0.88	2.30			0.99	4.61

Note: Values are provided for low and high quality epitaxial materials with optimized optical coatings.

Specific values for μ and τ are provided in Tables 9 and 10 as examples for low and high quality materials. A low quality material with a charge carrier mobility of $200 \text{ cm}^2/\text{Vs}$ and a minority carrier lifetime of 0.5 ns, for example, will have a minority carrier diffusion length of $0.5 \mu\text{m}$. A high quality material with a charge carrier mobility of $800 \text{ cm}^2/\text{Vs}$ and a minority carrier lifetime of 10 ns, will have a minority carrier diffusion length of $4.5 \mu\text{m}$.

TPV device power conversion efficiency is defined as electrical power generated divided by optical power absorbed by the TPV device. Optical power not absorbed by the TPV device, which includes reflection from the surface and below bandgap reflection by a back reflector, is assumed to return to the radiation source and is thus not considered a loss for efficiency calculations. Absorbed optical power includes absorption of photons by direct interband electronic transitions in the MQW material and by indirect phonon-assisted intervalley electronic transitions in the conduction and valence bands of n-type and p-type materials, respectively. An estimate for intervalley absorption can be made using the measured absorption coefficients for low energy photons in n-type GaSb where the primary photon absorption mechanism is by indirect phonon-assisted Γ -valley to L-valley electronic transitions,³² which are similar to the normal-to-oblique intervalley electronic transitions in IV–VI semiconductor MQW materials. With the same absorption mechanism in both the conduction and valence bands in IV–VI semiconductors, the same absorption coefficients for n-type and p-type materials can be used. (This is not the case for p-type III–V semiconductors where the dominant photon absorption mechanism in the valence band is direct intervalence band transitions between the light, heavy, and split-off bands near the Γ point, an effect that makes p-type III–V semiconductors much more absorptive of low energy photons.³³ GaSb p-type samples, when compared to n-type samples with comparable charge carrier concentrations, have absorption coefficients that are three times

larger at 500 meV and 10 times larger at 150 meV.³²) Measurements of n-type GaSb with a $2.2 \times 10^{17} \text{ cm}^{-3}$ electron concentration showed absorption coefficients increased from 10 cm^{-1} at 443 meV to 25 cm^{-1} at 150 meV, a spectral region that covers 90% of the 16.3 W/cm^2 of radiation below 443 meV from a 1400°C radiator with an emissivity of 0.9. Absorption coefficients for n-type GaSb with a $1.3 \times 10^{18} \text{ cm}^{-3}$ electron concentration increased from 50 to 150 cm^{-1} over the same spectral range. Assuming an absorption coefficient of 20 cm^{-1} for $8 \mu\text{m}$ of low charge carrier density IV–VI semiconductor MQW material and 150 cm^{-1} for $0.4 \mu\text{m}$ of high charge carrier density tunnel junction material gives a total absorption of 2.2%. A TPV device with a 443 meV bottom junction bandgap and nonoptimized optical coatings will have 8.2 W/cm^2 of below bandgap radiation transmitted into the device material, of which 179 mW/cm^2 will be dissipated as heat. When added to the 10.8 W/cm^2 of absorbed optical power within the MQW subbands, the total absorbed power increases by 1.7%. This estimate should be considered an upper limit for this below bandgap absorption effect since the assumed absorption coefficients are in the high range for the GaSb measurements and a generous thickness for the two tunnel junction is assumed. This optical absorption loss, being small, especially for devices with optimized doping and layer thicknesses, is not considered in the power conversion efficiencies discussed below.

Data for efficiency calculations can be obtained from Tables 4, 9 and 10. For example, a device with low quality material and nonoptimized optical coatings will generate 1.72 W/cm^2 of electrical power for a power conversion efficiency of 15.9%, while a high quality material will generate 3.86 W/cm^2 of electrical power for a power conversion efficiency of 35.7%. There will be more electrical power generated by devices with optimized optical coatings, 2.30 W/cm^2 and 4.61 W/cm^2 , for low and high quality materials, respectively, but with more optical power absorbed, 15.7 W/cm^2 , power conversion

efficiencies are lower, 14.6% and 29.4%, respectively. Absorption of more optical power reduces power conversion efficiency since there is more heat dissipation in the TPV device, which does not occur if incident optical power is reflected back to the radiator.

The photogenerated charge collection model described here shows that short circuit current densities with a 1400°C radiator are expected to be in the range of 2.3 to 5.3 A/cm² depending on material quality and optical coatings. As shown previously,¹⁷ open circuit voltage for small bandgap materials can be a large fraction of E_g/q when short circuit densities are large. For example, a 450 meV bandgap material will have an open circuit voltage of 182.3 mV at a short circuit current density of 0.5 A/cm², while at a short circuit current density of 5 A/cm² open circuit voltage increases more than 30% to 241.8 mV. As described in Khodr et al.,¹⁷ this effect can make it worthwhile to incorporate small bandgap materials into TPV device designs when large short circuit densities are possible.

Minority charge carrier lifetime is a key materials parameter that determines TPV device performance. As with short circuit current density, there is a strong relationship between open circuit voltage and minority charge carrier lifetime.¹⁷ For example, a 450 meV bandgap low quality material with a minority charge carrier lifetime of 0.1 ns will have an open circuit voltage of 173 mV, while a high quality material with a minority charge carrier lifetime of 10 ns will have an open circuit voltage of 233 mV. The open circuit voltages of high quality IV–VI semiconductor materials under high optical power conditions are in the same range as those demonstrated for high quality III–V semiconductor materials with similar bandgap energies and test conditions. For example, a 730 meV bandgap GaSb TPV device exposed to a SiC radiator at 1380°C exhibited a J_{sc} of 3.92 A/cm² and a V_{oc} of 0.44 V.³⁴ By comparison, our model predicts, for a similar bandgap 707 meV top junction material, J_{sc} values between 2.3 and 5.3 A/cm² and V_{oc} values between 0.39 V and 0.43 V. Open circuit voltage has a strong dependence on intrinsic charge carrier density, which is determined by the effective density of states in the conduction and valence bands of the MQW material. The (111)-oriented quantum confinement in MQW materials removes L-valley degeneracy and gives IV–VI semiconductors an intrinsic charge carrier density similar to that for a III–V semiconductor like GaSb. A good agreement between these modeling predictions and the results from experimental measurement of a similar bandgap material with a similar conduction band structure under similar operating conditions provides validation for the IV–VI semiconductor TPV device model presented here.

8 | COST MODEL

A molten silicon thermal energy storage system will be relatively low cost because its high latent heat energy density allows relatively small storage vessels requiring less insulation, which can be the most costly portion of a thermal energy storage vessel. Using cost estimates provided by Amy et al.,³⁵ the total materials cost for a molten silicon thermal energy storage system is estimated to be in the range of \$6.75 per kWh of energy storage capacity. Using this cost and assuming a round-trip efficiency of 35.7%, 10 h of power dispatched per day, and a 30 year lifetime, the energy storage hardware portion of the system will contribute \$0.0017/kWh to the LCOE. If power conversion efficiency is less because of low quality TPV device material, then a larger amount of thermal energy storage material would be needed. For example, energy storage hardware will contribute \$0.0039/kWh to the LCOE for a system with 15.9% power conversion efficiency.

Costs for TPV power conversion hardware include the TPV device, TPV device cooling hardware, and power inverter electronics. Cooling hardware cost is estimated to be \$0.08/W, and power inverter electronics cost is estimated to be \$0.07/W.³⁵ TPV device cost is primarily the cost of epitaxial layer growth, which scales with thickness. Based on a TPV device fabrication cost analysis that assumes reuse of costly III–V semiconductor growth substrates, the cost for epitaxial growth of III–V semiconductor materials has been estimated to be \$1/cm².³⁵ The optimal III–V semiconductor layer thickness for a 1400°C radiator is in the range of 8 μm,³⁶ which is the same thickness for an optimized triple junction IV–VI semiconductor TPV device fabricated from high quality material, see Tables 7 and 8. Assuming a \$1/cm² manufacturing cost, a IV–VI semiconductor TPV device made from high quality MQW material will have a cost between \$0.22/W and \$0.26/W depending on optical coating effectiveness. In the case of TPV devices made from low quality material, which perform optimally with thinner layers that match shorter minority carrier diffusion lengths, TPV device manufacturing cost will be less. A TPV device without optimized optical coatings and fabricated from low quality material, for example, will have a thickness of 3.36 μm, less than half of the 7.9 μm thickness of a TPV device made from high quality material. Assuming a \$0.50/cm² cost for TPV devices fabricated with thinner MBE layers gives an estimated cost between \$0.22/W and \$0.29/W, depending on optical coating effectiveness.

Table 11 provides example calculations for the contribution of TPV power conversion to the LCOE for a thermal energy storage system. TPV power conversion

TABLE 11 Levelized cost of energy (LCOE) calculations for molten silicon thermal energy storage with IV–VI semiconductor TPV devices fabricated from low and high quality materials and with two different operational lifetimes.

Material quality	TPV cost (\$/W)	Power conversion efficiency (%)	Operational lifetime (Years)	TPV cost over lifetime (\$/kWh)	Purchased energy cost (\$/kWh)	Dispatched energy cost (\$/kWh)	TPV and energy cost (\$/kWh)	Thermal storage cost (\$/kWh)	LCOE (\$/kWh)
Low	0.40	15.9	10	0.011	0.026	0.163	0.174	0.004	0.178
			3	0.036					0.203
High		35.7	10	0.011		0.073	0.084	0.002	0.086
			3	0.036					0.111

Note: TPV cost includes the TPV device, cooling, and inverter costs. Purchased energy cost is from Lazard³⁷ and thermal storage cost is from Amy et al.³⁵

cost for dispatched energy from a thermal energy storage system is the cost of TPV power conversion hardware plus purchased energy cost divided by power conversion efficiency. For example, TPV power conversion hardware costing \$0.40/W (assuming \$0.25/W for the TPV device plus \$0.15/W for device cooling and inverter electronics) with a 10 year service life will contribute \$0.011/kWh to the LCOE, while a service life of 3 years will contribute \$0.036/kWh. Assuming a purchased energy cost of \$0.026/kWh, which is the current low-range cost of wind power,³⁷ the marginal cost of dispatched energy is \$0.073/kWh for TPV devices with 35.7% power conversion efficiency and \$0.163/kWh for TPV devices with 15.9% power conversion efficiency. Adding TPV and thermal energy storage hardware costs gives LCOE values ranging from \$0.086/kWh to \$0.203/kWh. Operational and maintenance costs are expected to be negligible with a TPV-based latent heat energy storage system. It remains at a constant temperature during charge and discharge cycles, and the only moving parts in the entire energy storage system are for pumps that circulate heat exchange fluids. These LCOE calculations assume a purchased energy cost of \$0.026/kWh. Lower purchased energy costs can significantly reduce these LCOE values. For example, if energy can be purchased at \$0.013/kWh, which is predicted for future utility-scale solar photovoltaic systems,³⁸ then a TPV device with 15.9% power conversion efficiency, and a 3-year operational lifetime will give a thermal energy storage LCOE of \$0.12/kWh. This cost analysis shows that there will be economic viability even for TPV devices that are not fully optimized in terms of power conversion efficiency and operational lifetime.

9 | DISCUSSION

The ability to grow high crystalline quality narrow bandgap IV–VI semiconductor materials on low cost silicon substrates is the key technological capability that

enables development of the low cost energy storage technology described above. Uniformly high quality IV–VI semiconductor material, as assessed by PL intensity, X-ray diffraction, and Hall effect measurements, across entire three-inch diameter (111)-oriented silicon wafers can be typically achieved with the appropriate MBE growth conditions, which include good oxide desorption and growth of a thin CaF₂ layer before IV–VI semiconductor layer growth, as described previously.⁶ Unpublished Hall effect measurements have shown that room temperature electron and hole mobilities are as high as 1500 cm²/Vs for Pb_{1-x}Sn_xSe epilayers with a 10% tin content on the group IV sublattice, while they decrease to about 200 cm²/Vs for Pb_{1-x}Sr_xSe epilayers with a 20% strontium content on the group IV sublattice. The 800 cm²/s charge carrier mobility assumed for high quality MQW materials is near the average of these two mobilities, so there is experimental support for this assumption. There is also experimental support for the assumption of a 10 ns minority carrier lifetime for high quality material. For example, time-resolved PL measurements of PbSe epilayers grown on thermal expansion mismatched (111)-oriented InAs substrates showed minority carrier lifetimes of 20 ns,³⁹ and measurements of optical pumping lasing thresholds for PbSrS/PbS/PbSrS heterostructures grown on (111)-oriented silicon showed minority carrier lifetimes of 10 ns.¹¹ Therefore, the predictions for electrical power generation shown in Tables 9 and 10 should be considered realistic estimates for the performance of TPV devices made from these materials.

A TPV device fabricated with high quality material and without optimized optical coatings when exposed to a 1400°C radiator will absorb 10.8 W/cm² of optical power, of which 3.9 W/cm² will be converted to electrical power leaving 6.9 W/cm² of heat generation in the TPV device material. Most of this heat will be generated in the top junction material where the median absorbed photon

energy is 196 meV greater than the bandgap energy resulting in a large amount of lattice heating due to phonon generation during charge carrier thermal relaxation. Heating effects in IV–VI semiconductor thin films on (111) silicon substrates have been studied during PL experiments where near-IR laser illumination provides a similar amount of heat flux. For example, PL measurements with a 4.2 μm thick IV–VI semiconductor layer on silicon at 25°C showed an epilayer surface temperature of 83.5°C for an incident optical power density of 28.7 W/cm².⁴⁰ Optical power measurements showed 41% reflection of 45° incident near-IR laser power from uncoated IV–VI semiconductor layers,^{41,42} so the heat input flux in these experiments was in the range of 16.9 W/cm², more than twice as large as the expected heat input flux for TPV device operation with 1400°C radiators. Experiments involving repeated PL measurement of the same sample from an MBE-grown PbSrSe/PbSe/PbSrSe MQW material on silicon showed no visible sample deterioration or decrease in PL emission intensity.⁹ Ability to tolerate thermal cycling strains when epilayer surfaces are repeatedly heated from room temperature to 83.5°C suggest that they will be reliable when used as TPV device materials in a thermal energy storage system. As discussed above, the same easy dislocation glide system that allows (111)-oriented IV–VI semiconductor thin films to deform plastically following MBE growth at 350°C also allows them to maintain high crystalline quality when used in a high temperature application. The TPV materials and device models presented here can be used generally to optimize layer thicknesses based on measured material parameters (charge carrier mobility from Hall effect measurements and minority charge carrier lifetime from reverse bias saturation current measurements) and predict device performance for different TPV power conversion applications. The device model here can be refined to incorporate the effects of a nonzero series resistance, which is likely for wider bandgap MQW materials where donor and acceptor doping can be less effective. Future device fabrication work should therefore focus on doping optimization of wider bandgap materials. This work can include investigation of PbEuSe alloys,¹⁸ which have a similar bandgap energy dependence on alloy composition as PbSrSe, for use as the barrier layer in MQW materials. Device fabrication will require patterned etching and metallization of MBE-grown material. Of particular concern is the possibility of creating surface states on etched IV–VI semiconductor sidewalls since this could make surface recombination nonnegligible. The next step in IV–VI semiconductor TPV device development is to fabricate etched and metallized layer structures and demonstrate TPV power generation from fabricated

devices. Device performance parameters can be used to determine materials quality parameters, which can then be used with the device model to optimize layer thicknesses.

Although this paper highlighted a specific thermal energy storage application with a 1400°C radiator, there are many other applications such as in combined heat and power (CHP) where lower radiator temperatures will likely be involved. The MQW materials design model described in this paper offers flexibility in obtaining semiconductor materials, grown using manufacturing-friendly MBE growth methods, that will have optimal bandgaps for many different TPV device applications. Both the materials and device design models can also be used to understand the effects of lattice heating. The data in Tables 9 and 10 are for a TPV device with a lattice temperature of 25°C. The effects of a hotter lattice temperature, for example, are shown through Equations (17) and (18) and the intrinsic carrier density. At a lattice temperature of 52°C (325 K), which is expected to be the case for a TPV device exposed to a 1400°C radiator, the reverse bias saturation current increases by a factor of 4 in the bottom junction and by more than a factor of 8 in the top junction. These increases, which are due to the increased thermal generation of electron-hole pairs, reduce the open circuit voltage and power density by about 8%. Future modeling efforts can be built upon the basic models presented here. For example, finite element thermal modeling can be performed to assess TPV device performance for different thermal management approaches. Results from this future modeling work will help to guide the development of optimized TPV materials and devices for power conversion applications.

MBE growth of IV–VI semiconductor layers on silicon typically involves the use of a thin CaF₂ buffer layer.⁶ An insulating CaF₂ layer under the TPV device material is attractive since it allows the complete electrical isolation of etched IV–VI semiconductor structures. Photolithographically patterned processing steps involving a mesa isolation etch down to the CaF₂ buffer layer, a contact via etch down to the bottom p-type layer, insulator deposition, contact via etches through the insulator to the n-type and p-type layers, and metal deposition allow fabrication of series connected TPV devices from these materials. Such monolithic series connection has been demonstrated for TPV devices fabricated from narrow bandgap III–V semiconductor materials.⁴³ In this case, however, additional processing steps involving removal of the GaSb growth substrate and transferring the epilayer to a semi-insulating GaAs handle wafer were needed so that fully electrically isolated devices could be obtained. On the other hand, CaF₂-coated silicon growth substrates used for IV–VI semiconductor MQW materials growth allow chip-

level series interconnection without an additional epitaxial layer transfer processing step. Simpler series-connected device processing and ability to use existing silicon manufacturing infrastructure makes IV–VI semiconductor on silicon TPV materials technology attractive for rapid scaling of manufacturing capacity. The benefits of a low cost silicon growth substrate technology can be seen with the cost model outlined above. An estimate for the at-scale manufacturing cost contribution of the silicon substrate to the LCOE for TPV power conversion can be obtained from the current retail cost of crystalline silicon solar panels, which are mostly silicon and cost about \$0.01/cm². Assuming a TPV device is made from high quality material and a 10-year operational lifetime, this cost contributes only about \$0.0012/kWh to the LCOE. By contrast, a similarly performing III–V semiconductor TPV device fabricated using GaSb growth substrates, which currently cost more than \$20/cm², will have a cost contribution of more than \$2.40/kWh to the LCOE, and this is too expensive for economic viability. Development of a lower cost TPV device materials technology, as described in this paper, will help to eliminate a barrier to commercial adoption of a thermal-to-electrical power conversion technology. Future work in this area promises to offer a pathway for the development and deployment of low cost thermal energy storage technology.

10 | SUMMARY

This paper described how two IV–VI semiconductor alloys, Pb_{0.81}Sn_{0.19}Se and Pb_{0.8}Sr_{0.2}Se, can be used as the barrier and well materials, respectively, for MQW materials that absorb mid-infrared radiation. Bandgap energy as a function of Pb_{0.81}Sn_{0.19}Se well width calculations included the effects of epitaxial layer strain. A specific example involving MQW materials with well widths of 1.4, 2.45, and 3.5 nm, and bandgap energies of 707, 559, and 443 meV, respectively, for triple junction TPV device fabrication was provided. A diffusion-only charge collection model was used to obtain analytical expressions for internal quantum efficiencies on the n-type and p-type sides of each junction. Optimal layer thicknesses for each layer at a pn junction TPV device structure were found using these expressions. The power generation levels with a 1400°C radiator for current-matched junctions in a triple junction TPV device were predicted to be 1.7 W/cm² for low quality MQW materials and 4.6 W/cm² for high quality MQW materials. A cost model for TPV device power conversion from a molten silicon thermal energy storage

system predicted LCOE values that are competitive with other energy storage and power generation technologies. Cost calculations also showed that TPV devices, even ones fabricated from low quality materials, can provide energy at competitive costs (Table 6).

ORCID

Patrick J. McCann  <http://orcid.org/0000-0002-9854-2005>

Majed Khodr  <http://orcid.org/0000-0001-6149-5834>

REFERENCES

1. Chubb DL, Good BS, Lowe RA. Solar thermophotovoltaic (STPV) system with thermal energy storage. *AIP Conf Proc*. 1996;358:181-198.
2. Datas A. Optimum semiconductor bandgaps in single junction and multijunction thermophotovoltaic converters. *Sol Energy Mater Sol Cells*. 2015;134:275-290.
3. Müller P, Zogg H, Fach A, et al. Reduction of threading dislocation densities in heavily lattice mismatched PbSe on Si (111) by glide. *Phys Rev Lett*. 1997;78:3007-3010.
4. Wu HZ, Fang XM, Salas R, McAlister Jr. D, McCann PJ. Molecular beam epitaxy growth of PbSe on BaF[sub 2]-coated Si(111) and observation of the PbSe growth interface. *J Vacuum Sci Technol B: Microelectr Nanometer Struct*. 1999;17:1263.
5. Zogg H, Blunier S, Fach A, et al. Thermal-mismatch-strain relaxation in epitaxial CaF₂, BaF₂/CaF₂, and PbSe/BaF₂/CaF₂ layers on Si(111) after many temperature cycles. *Phys Rev B: Condens Matter Mater Phys*. 1994;50:10801-10810.
6. McCann PJ, Fang XM, Liu WK, Strecker BN, Santos MB. MBE growth of PbSe/CaF₂/Si(111) heterostructures. *J Cryst Grow*. 1997;175/176:1057-1062.
7. McAlister DW, McCann PJ, Namjou K, Fang XM, Alkhouli O, Wu HZ. Midinfrared photoluminescence from IV–VI semiconductors grown on silicon. *J Appl Phys*. 2001; 89:3514-3516.
8. Wu HZ, McCann PJ, Alkhouli O, et al. Molecular beam epitaxial growth of IV–VI multiple quantum well structures on Si(111) and BaF[sub 2](111) and optical studies of epilayer heating. *J Vacuum Sci Technol B: Microelectron Nanometer Struct*. 2001;19:1447.
9. Elizondo LA, Li Y, Sow A, et al. Optically pumped mid-infrared light emitter on silicon. *J Appl Phys*. 2007;101:104504.
10. Zogg H, Alchalabi K, Zimin D, Kellermann K. Lead chalcogenide on silicon infrared sensors: focal plane array with 96×128 pixels on active Si-chip. *Infrared Phys Technol*. 2002;43:251-255.
11. Khair A, Rahim M, Fill M, et al. Modular PbSrS/PbS mid-infrared vertical external cavity surface emitting laser on Si. *J Appl Phys*. 2011;110:023101.
12. Wiesauer K, Springholz G. Critical thickness and strain relaxation in high misfit heteroepitaxial systems: PbTe_{1-x}Se_x on PbSe (001). *Phys Rev B: Condens Matter Mater Phys*. 2004;69:245313.
13. Zasavitskii II, de Andrada e Silva EA, Abramof E, McCann PJ. Optical deformation potentials for PbSe and PbTe. *Phys Rev B: Condens Matter Mater Phys*. 2004;70:115302.

14. Preier H. Recent advances in lead-chalcogenide diode lasers. *Appl Phys*. 1979;20:189-206.
15. Shen WZ, Yang HF, Jiang LF, et al. Band gaps, effective masses and refractive indices of PbSrSe thin films: key properties for mid-infrared optoelectronic device applications. *J Appl Phys*. 2002;91:192.
16. Wu HZ, Dai N, Johnson MB, McCann PJ, Shi ZS. Unambiguous observation of subband transitions from longitudinal valley and oblique valleys in IV–VI multiple quantum wells. *Appl Phys Lett*. 2001;78:2199-2201.
17. Khodr M, Chakraborty M, McCann PJ. PbSnSe/PbSrSe quantum well materials for thermophotovoltaic devices. *AIP Adv*. 2019;9:035303. doi:10.1063/1.5080444
18. Fang XM, I-Na Chao, BN, Strecker, PJ, et al. Molecular beam epitaxial growth of Bi₂Se₃- and Tl₂Se-doped PbSe and PbEuSe on CaF₂/Si(111). *J Vacuum Sci Technol B: Microelectron Nanometer Struct*. 1998;16:1459.
19. Norton P, Tacke M. MBE of Pb_{1-x}Eu_xSe for the use in IR devices. *J Cryst Grow*. 1987;81:405-410.
20. Wang H, Pei Y, LaLonde AD, Snyder GJ. Heavily doped p-type PbSe with high thermoelectric performance: an alternative for PbTe. *Adv Mater*. 2011;23:1366-1370.
21. Parada NJ, Pratt GW. New model for vacancy states in PbTe. *Phys Rev Lett*. 1969;22:180-182.
22. McCann PJ, Li L, Furneaux JE, Wright R. Optical properties of ternary and quaternary IV–VI semiconductor layers on (100) BaF₂ substrates. *Appl Phys Lett*. 1995;66:1355-1357.
23. Khodr MF, McCann PJ, Mason BA. Effects of band nonparabolicity on the gain and current density in EuSe-PbSe/sub 0.78/Te/sub 0.22/-EuSe IV–VI semiconductor quantum-well lasers. *IEEE J Quantum Electron*. 1996;32:236-247.
24. Findlay PC, Pidgeon CR, Kotitschke R, et al. Auger recombination dynamics of lead salts under picosecond free electron laser excitation. *Phys Rev B: Condens Matter Mater Phys*. 1998;58(12):12908-12915.
25. Sachar HK, Chao I, McCann PJ, Fang XM. Growth and characterization of PbSe and Pb_{1-x}Sn_xSe on Si(100). *J Appl Phys*. 1999;85:7398-7403.
26. Chan WR, Bermel P, Pilawa-Podgurski RCN, et al. Toward high-energy-density, high-efficiency, and moderate-temperature chip-scale thermophotovoltaics. *Proc Natl Acad Sci USA*. 2013;110:5309-5314.
27. Weng B, Qiu J, Yuan Z, Larson PR, Strout GW, Shi Z. Responsivity enhancement of mid-infrared PbSe detectors using CaF₂ nano-structured antireflective coatings. *Appl Phys Lett*. 2014;104:021109. doi:10.1063/1.4861186
28. Ghebrehirhan M, Bermel P, Yeng YX, Celanovic I, Soljačić M, Joannopoulos JD. Tailoring thermal emission via Q matching of photonic crystal resonances. *Phys Rev A*. 2011;83:033810.
29. Li Z, Pestourie R, Park JS, Huang YW, Johnson SG, Capasso F. Inverse design enables large-scale high-performance meta-optics reshaping virtual reality. *Nat Commun*. 2022;13:2409. doi:10.1038/s41467-022-29973-3
30. Sinkkonen J, Ruokolainen J, Uotila P, Hovinen A. Spatial collection efficiency of a solar cell. *Appl Phys Lett*. 1995;66:206-208.
31. Green MA. Accuracy of analytical expressions for solar cell fill factors. *Sol Cells*. 1982;7:337-340.
32. Chandola A, Pino R, Dutta PS. Below bandgap optical absorption in tellurium-doped GaSb. *Semicond Sci Technol*. 2005;20:886-893. doi:10.1088/0268-1242/20/8/046
33. Casey Jr. HC, Sell DD, Wecht KW. Concentration dependence of the absorption coefficient for n- and p-type GaAs between 1.3 and 1.6 eV. *J Appl Phys*. 1975;46:250. doi:10.1063/1.321330
34. Burger T, Sempere C, Roy-Layinde B, Lenert A. Present efficiencies and future opportunities in thermophotovoltaics. *Joule*. 2020;4:1660-1680.
35. Amy C, Seyf HR, Steiner MA, Friedman DJ, Henry A. Thermal energy grid storage using multi-junction photovoltaics. *Energy Environ Sci*. 2019;12:334.
36. Gamel MMA, Ker PJ, Lee HJ, et al. Multi-dimensional optimization of In_{0.53}Ga_{0.47}As thermophotovoltaic cell using real coded genetic algorithm. *Sci Rep*. 2021;11:7741. doi:10.1038/s41598-021-86175-5
37. Lazard. *Levelized Cost of Energy, Levelized Cost of Storage, and Levelized Cost of Hydrogen*. Lazard report, October 28, 2021. Accessed February 7, 2022. <https://www.lazard.com/perspective/levelized-cost-of-energy-levelized-cost-of-storage-and-levelized-cost-of-hydrogen/>
38. Bolinger M, Wiser R, O'Shaughnessy E. Levelized cost-based learning analysis of utility-scale wind and solar in the United States. *iScience*. 2022;25:104378.
39. Haidet BB, Nordin L, Muhowski AJ, et al. Interface structure and luminescence properties of epitaxial PbSe films on InAs(111)A. *J Vac Sci Technol A*. 2021;39:023404. doi:10.1116/6.0000774
40. Li YF, McCann PJ, Sow A, Yao C, Kamat PC. Improvement in heat dissipation by transfer of IV–VI epilayers from silicon to copper. *IEEE Photonics Technol Lett*. 2004;16:2433-2435.
41. Jeffers JD, Namjou K, Cai Z, McCann PJ, Olona L. Cross-plane thermal conductivity of a PbSnSe/PbSe superlattice material. *Appl Phys Lett*. 2011;99:041903.
42. McCann PJ, Olona L, Cai Z, Jeffers JD, Namjou K. MBE-grown IV–VI semiconductor structures for thermal conductivity measurements. *MRS Proc*. 2012;1404:mrsf11-1404-w04-05.
43. Wang CA, Huang RK, Shiao DA, et al. Monolithically series-interconnected GaInAsSb/AlGaAsSb/GaSb thermophotovoltaic devices with an internal backsurface reflector formed by wafer bonding. *Appl Phys Lett*. 2003;83:1286-1288.

How to cite this article: McCann PJ, Khodr M. Materials and device designs for thermophotovoltaic power conversion. *Energy Sci Eng*. 2023;11:2469-2485. doi:10.1002/ese3.1467